Integrating Rational Functions With Partial Fractions

Splitting a rational function into distinct linear pieces, evaluating a definite integral after the split, and taking an improper integral to a limit

The one move behind every problem. A rational function whose denominator factors into *distinct linear* factors is a sum of constants over those factors, and each piece integrates to a logarithm: $\dfrac{N(x)}{(x-r_1)(x-r_2)} = \dfrac{A}{x-r_1} + \dfrac{B}{x-r_2}$.

Directions.

- Routine: factor the denominator, write a constant over each factor, clear denominators, then substitute each root in turn — every term but one vanishes.
- Check the degree first: if the numerator's degree is not lower than the denominator's, divide before you split.
- Show the decomposition before the integration, keep absolute values inside every logarithm, and include $+C$ on indefinite integrals.
- Give exact answers, and combine a difference of logarithms into one logarithm.

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1. $\displaystyle\int \frac{5}{(x-2)(x+3)}\,dx.$

(a) Write the decomposition, clear denominators, and find A and B by substituting the two roots. **(b)** Integrate.

Answer: ________________

2. $\displaystyle\int \frac{5x+1}{(x-1)(x+2)}\,dx.$

(a) The numerator is linear now. Say in one sentence why the same two-constant form still works, then find A and B. **(b)** Integrate.

Answer: ________________

3. Marginal cost. A workshop's marginal cost at production level x (hundreds of units) is $C'(x) = \dfrac{24}{(x+1)(x+3)}$ dollars per unit.

(a) Decompose $C'(x)$. **(b)** The added cost of going from $x = 0$ to $x = 1$ is $\displaystyle\int_0^1 C'(x)\,dx$; evaluate it exactly and combine into one logarithm. **(c)** State the units and say what the number measures.

Answer: ________________

4. $\displaystyle\int \frac{2}{x(x-1)(x+1)}\,dx.$

(a) Three distinct factors need three constants. Clear denominators and substitute $x = 0$, $x = 1$, $x = -1$. **(b)** Integrate, and say why the root substitution made one constant negative.

Answer: ____________________

5. Filtration. Water passes through a filter at the rate $r(t) = \dfrac{12}{(t+1)(t+3)}$ litres per minute, t minutes after start-up.

(a) Decompose $r(t)$. **(b)** Evaluate $\displaystyle\int_0^3 r(t)\,dt$ exactly and combine into one logarithm. **(c)** Is the rate increasing or decreasing on this interval? Justify from the formula, not from a graph.

Answer: ____________________

6. **Leak.** A tank leaks a chemical at the rate $r(t) = \dfrac{2}{t(t+2)}$ kilograms per hour, for $t \geq 1$. **(a)** Decompose $r(t)$ and write the antiderivative as a single logarithm. **(b)** Evaluate it on $[1, b]$ and take the limit as $b \to \infty$. **(c)** Does $\displaystyle\int_1^\infty r(t)\,dt$ converge or diverge? Give its value, and put your part (b) limit and part (c) value on the answer line.

Answer: ____________________

7. Logistic growth. A population p (thousands) satisfies $\dfrac{dp}{dt} = \dfrac{p(10-p)}{10}$, so the time to grow from $p = 2$ to $p = 4$ is $\displaystyle\int_2^4 \frac{10}{p(10-p)}\, dp$ years.
(a) Decompose the integrand. Both constants are positive — say why the $10 - p$ factor does not make its own constant negative. **(b)** Evaluate the integral exactly, as one logarithm.
(c) Without computing it, explain why growing from $p = 4$ to $p = 6$ takes less time.

Answer: __________________

8. Challenge — total emission. A process releases a pollutant at the rate $r(t) = \dfrac{6}{(t+1)(t+4)}$ kilograms per year, $t \geq 0$.
(a) Decompose $r(t)$ and write the antiderivative as a single logarithm. **(b)** Rewrite the total emission as $\lim_{b\to\infty} \int_0^b r(t)\,dt$, evaluate on $[0, b]$, and take the limit. **(c)** Does it converge? Give the exact total, then explain why $r(t) \to 0$ is not on its own enough — name a rate that decays to 0 and still gives an infinite total.

Answer: ________________